

VENKATESH SWARNA

Data Engineer

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Professional Summary

Results-driven Data Engineer with 5 years of experience in designing, building, and optimizing scalable data pipelines, ETL workflows, and modern data architectures across Data, ML, and AI domains. Proficient in Python, SQL, Apache Spark, AWS, and Azure, with hands-on exposure to machine learning workflows, MLOps, and AI integration. Skilled in data modeling, warehousing, and real-time processing, delivering end-to-end data solutions that support business intelligence and advanced analytics. Adept at collaborating with cross-functional teams to translate complex business requirements into actionable data-driven insights. Passionate about leveraging GenAI, cloud-native tools, and advanced analytics to drive innovation and empower organizational decision-making.

Skills

Programming & Scripting: Python (NumPy, Pandas, OOP), SQL, PySpark, R (basic), Shell Scripting

Big Data & Data Engineering: Apache Spark (batch & streaming), Hadoop, Hive, HDFS, Data Modeling, Star/Snowflake Schemas, Data Warehousing (OLAP/OLTP)

Cloud Platforms: AWS (S3, Glue, Redshift, Lambda, Kinesis, EMR, DynamoDB), Azure (Data Factory, Databricks, Synapse, Data Lake, Cosmos DB), Google Cloud (BigQuery)

Databases: SQL Server, PostgreSQL, MySQL, Oracle, Teradata, MongoDB, NoSQL

ETL & Workflow Orchestration: Apache Airflow, Talend, Informatica PowerCenter, SSIS, SSRS, AWS Glue, Azure Data Factory

Streaming & Real-Time Analytics: Apache Kafka, Apache Flink (overview), Spark Structured Streaming

BI & Visualization: Power BI, Tableau

Containerization & Orchestration: Docker, Kubernetes

Machine Learning & AI: scikit-learn, TensorFlow, GenAI integration (OpenAI API, LangChain), ML pipelines, Data preprocessing & feature engineering

Methodologies & Tools: Agile, Kanban, Data Analysis, Data Visualization, Data Mining, Data Normalization

Operating Systems: Linux, Windows

Education

University of New Haven, West Haven, CT
Masters, Data Science

Aug 2023 - May 2025

Acharya Nagarjuna University, Guntur, India
Bachelor of Technology, Electronics and Communication Engineering

Sep 2013 - Apr 2017

Professional Experience

Cognizant | Data Analyst – Programmer Analyst

Oct 2021 - Aug 2023

- Designed and implemented scalable, high-performance data transformation pipelines on Azure Data Factory and Databricks, adhering to medallion architecture (bronze, silver, gold) best practices in ADLS Gen2.
- Migrated and optimized ETL pipelines from SQL Server to Spark SQL on Databricks, ensuring compatibility and improving processing speed by 30%.
- Integrated data from 10+ sources (SQL, APIs, JSON, flat files) into Azure Data Lake, enabling centralized analytics and reducing data silos.
- Developed and optimized ETL workflows using AWS Glue, EMR, and Lambda, handling 2TB+ of data daily and achieving 25% cost savings through optimized resource allocation.
- Implemented real-time data streaming using Apache Kafka and AWS Kinesis, reducing processing latency by 20% for time-sensitive applications.
- Automated job scheduling, monitoring, and alerts using Apache Airflow, improving pipeline reliability by 40%.
- Partnered with data scientists to preprocess and structure datasets for ML models in TensorFlow, Scikit-learn, and Keras, resulting in a 15% increase in model accuracy.
- Partnered with QA teams for data validation, bug fixes, and data quality checks, improving overall integrity by 10%.
- Adhered to Agile/Kanban methodologies, achieving 100% sprint delivery success across multiple projects.
- Supported CI/CD integration of ADF and Databricks pipelines with GitHub, streamlining deployments and version control.
- Enhanced pipeline scalability by modularizing ETL components, reducing maintenance time for future deployments.

- Collaborated with business stakeholders to define KPIs and automate reporting dashboards, improving data-driven decision-making across departments.
- Documented data pipelines, workflows, and best practices, ensuring smooth knowledge transfer and reducing onboarding time for new team members by 35%.

Accenture | *Application Developer*

Dec 2018 - Oct 2021

- Designed and developed data pipelines in Azure Data Factory for data ingestion and orchestration from multiple enterprise sources.
- Implemented data transformation and aggregation workflows using Azure Databricks (PySpark + Spark SQL), ensuring high-quality, validated data for business reporting.
- Defined and maintained folder structures and data lake architecture on ADLS Gen2 following medallion (bronze, silver, gold) best practices.
- Automated data extraction and ingestion from emails using Power Automate (MS Flow) and stored data in SharePoint Lists, reducing manual entry and saving 15–20 hours weekly.
- Created real-time Power BI dashboards, improving reporting accuracy and enabling timely decision-making.
- Built ETL workflows with SSIS and SQL Server, streamlining enterprise reporting and audit compliance.
- Applied data quality checks, cleansing, and business rules leveraging reusable frameworks, improving overall accuracy of downstream reports.
- Collaborated with stakeholders and business teams (pricing, operations, audit) to gather requirements and deliver end-to-end BI solutions.
- Collaborated with QA teams to conduct unit and regression testing, ensuring high-quality deliverables and on-time completion of sprint goals within Agile frameworks.

Projects

Stock Market Price Analysis *Masters Project*

- Built a real-time streaming pipeline using Apache Kafka and Python, ingesting ~500 stock events per minute from Google Finance NASDAQ.
- Processed and stored ~2GB of daily stock data on AWS S3 using Glue and Athena, reducing query times from 10s to 3s.
- Developed dynamic Power BI dashboards with 10–15s refresh intervals, enabling near real-time trend analysis.
- Improved decision-making efficiency by ~25% through automated insights and faster reporting.
- Applied AWS services to design a scalable, cost-effective architecture for financial data analytics.

US Accidents Severity Prediction *Masters Project*

- Built and tuned Random Forest and XGBoost models on ~300K accident records, achieving 82–85% accuracy and a 0.78 F1-score.
- Conducted data cleaning, transformation, and feature engineering to improve dataset quality by ~15%.
- Implemented a lightweight prediction tool delivering results in under 30s, reducing manual analysis time by 50%.
- Created visual analytics to identify accident hotspots and contributing factors for proactive safety measures.
- Delivered actionable insights that supported transportation safety strategies using ML-based predictions.

Corrosion Classification *Masters Project*

- Designed and trained a YOLOv5x image classification model to detect types of corrosion in industrial assets.
- Collected, labeled, and managed datasets using Roboflow, ensuring high-quality annotations for training.
- Improved accuracy to 92% and F1-score to 0.90 through hyperparameter tuning and model optimization.
- Deployed model outputs to support predictive maintenance, reducing operational costs by minimizing downtime.
- Collaborated on dataset preparation, augmentation, and evaluation workflows for scalable model retraining.

Certifications & Achievements

Microsoft Azure Fundamentals (AZ-900)

AWS Certified Data Engineer - Associate (DEA-C01)

Star of the Month award for Couple of times during my tenure in Accenture.